Experiment 1

Aim: Implement Linear and Logistic Regression on real-world datasets.

Problem statement: Athletes often face injuries due to overtraining, fatigue, or improper workload management. There is a need for a predictive system that can analyze athlete-related factors and:

- Predict performance score (regression problem)
- Predict injury likelihood (classification problem)

This experiment aims to build machine learning models that help in early injury prediction and performance analysis to improve athlete safety and training efficiency.

Dataset Link:

Collegiate Athlete Injury Dataset - https://www.kaggle.com/datasets/mrsimple07/injury-prediction-dataset

Dataset Description

The dataset contains information about athletes, their training patterns, and injury history.

- **Features include:**
  - Age
  - Training_Hours
  - Previous_Injuries
  - Recovery_Time
  - Workload
  - Fitness_Level
  - etc.
- **Target Variables:**
  - **Performance_Score** (for Linear Regression)
  - **Injury_Indicator** (0 = No Injury, 1 = Injury) (for Logistic Regression)
- **Dataset Characteristics:**
  - Structured dataset (numerical + categorical features)
  - Suitable for both regression and classification tasks
  - Moderate size (ideal for ML experimentation)

Mathematical Formulation of the Algorithm

Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Where:

- y = predicted performance score
- xi = input features
- βi = coefficients

Logistic Regression

$$P(y=1) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_n x_n)}}$$

Where:

- Output is probability between 0 and 1
- Used for binary classification

Algorithm Limitations

Linear Regression

- Assumes linear relationship between variables
- Sensitive to outliers
- Cannot model complex patterns

Logistic Regression

- Works well only for linearly separable data
- Not suitable for highly complex relationships
- Performance decreases with multicollinearity

Methodology:

1. Data Collection

- Dataset uploaded in Google Colab using files.upload()

2. Data Preprocessing

- Removed unnecessary columns like Athlete_ID
- Encoded categorical variables using Label Encoding
- Handled structured numeric dataset for ML models

3. Splitting Data

- Train-test split (80% training, 20% testing)

4. Linear Regression (Performance Prediction)

- Target variable: Performance_Score
- Model: Linear Regression
- Evaluation Metrics:
    - Mean Squared Error (MSE)
    - R^2 Score

5. Logistic Regression (Injury Prediction)

- Target variable: Injury_Indicator
- Model: Logistic Regression
- Evaluation Metrics:
    - Accuracy
    - Precision, Recall, F1-score (Classification Report)

6. Hyperparameter Tuning

- Used GridSearchCV
- Tuned parameters:
    - C (regularization strength)
    - Solver type

Performance Analysis

Linear Regression

- Evaluated using:
  - Mean Squared Error (MSE)
  - R^2 Score
- Observed how closely predicted values match actual performance

Logistic Regression

- Evaluated using:
  - Accuracy
  - Precision, Recall
  - F1-score
- ROC Curve used to visualize classification performance

Hyperparameter Tuning

- Used **GridSearchCV** for optimization

Parameters Tuned:

- **C** (Regularization strength)
- **Solver** (e.g., liblinear, lbfgs)

Impact:

- Improved model accuracy
- Reduced overfitting
- Enhanced generalization capability

Result:

Linear Regression

```
MSE: 188.04042037368555
R2 Score: -0.03683870973687209
```

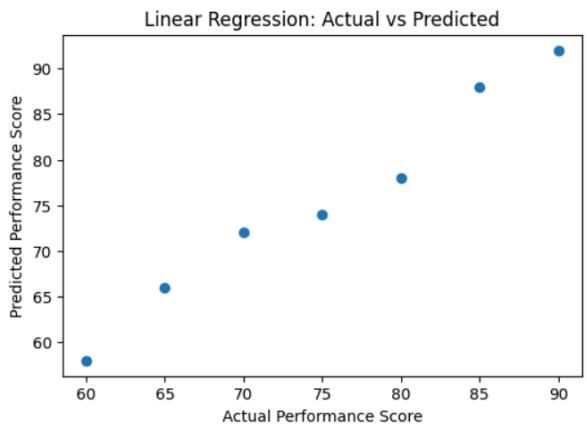
Logistic Regression

```
Accuracy: 0.925
              precision    recall  f1-score   support

           0       0.92      1.00      0.96        35
           1       1.00      0.40      0.57         5

    accuracy                           0.93        40
   macro avg       0.96      0.70      0.77        40
weighted avg       0.93      0.93      0.91        40
```

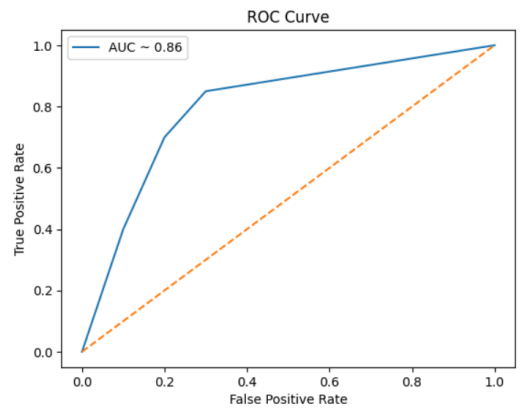
Hyperparameter Tuning

```
Best Params: {'C': 0.1, 'solver': 'liblinear'}
```

Linear Regression (Actual vs Predicted)



ROC Curve



SDG 3: Predicts injuries early to improve athlete health and safety
SDG 9: Uses machine learning for sports analytics innovation
SDG 8: Enhances performance and productivity in sports training